

L05 The Simplex Method

CS 169 / CS 268
Introduction to Optimization

Ioannis Panageas

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The idea of simplex

The **simplex method** searches among basic feasible solutions.

In a nondegenerate two-dimensional picture, a step moves along an edge to a neighboring vertex.

Choose an improving direction, move as far as feasibility allows, and repeat.

We will follow one example and explain why it stops.

Our running example

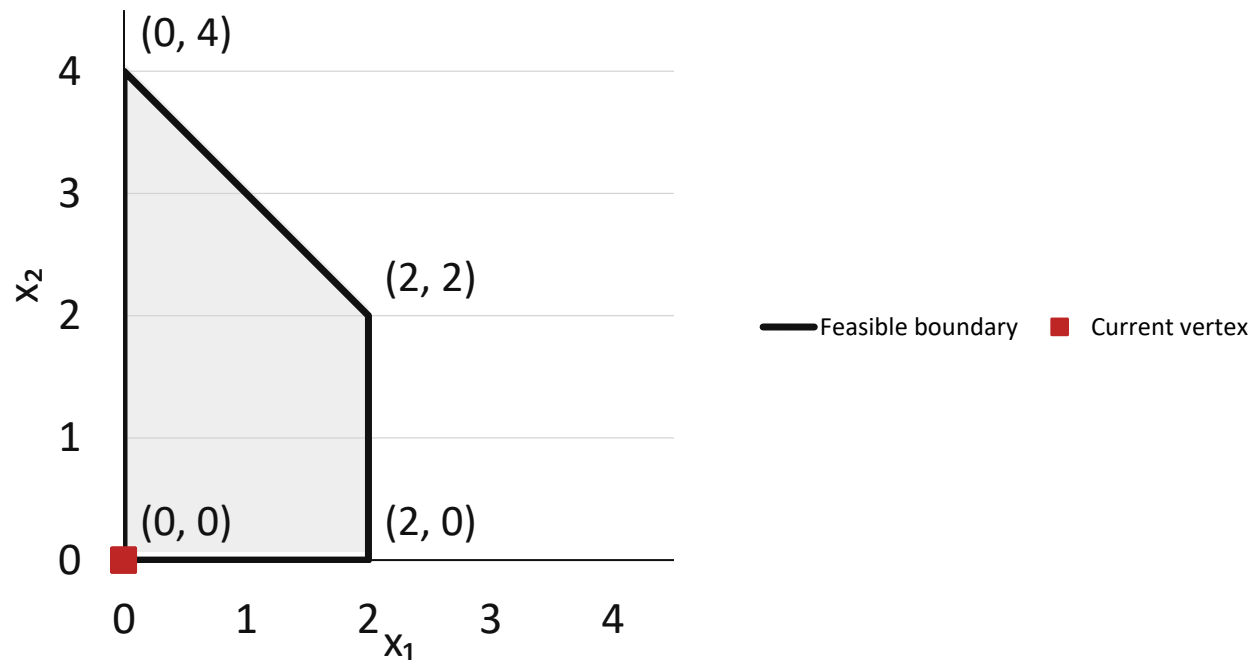
$$\begin{array}{ll}\text{maximize} & p = 3x_1 + 2x_2 \\ \text{subject to} & x_1 + x_2 \leq 4, \\ & x_1 \leq 2, \\ & x_1, x_2 \geq 0.\end{array}$$

Each unit uses one machine-hour. There are 4 hours, and at most 2 units of product 1 can be sold.

We keep the **maximization** form throughout the example.

The starting vertex

$$(x_1, x_2) = (0, 0), \quad p = 0.$$



Both coordinate directions initially increase profit.

Slacks measure unused capacity

$$s_1 = 4 - x_1 - x_2, \quad s_2 = 2 - x_1.$$

s_1 is unused machine time. s_2 is unused sales capacity for product 1.

$$x_1, x_2, s_1, s_2 \geq 0.$$

A slack reaches zero exactly when its constraint becomes tight.

A basic feasible solution

There are two independent equalities in four nonnegative variables.

$$x_1 + x_2 + s_1 = 4, \quad x_1 + s_2 = 2.$$

Set the **nonbasic variables** x_1, x_2 to zero and solve for the basic variables s_1, s_2 .

$$(x_1, x_2, s_1, s_2) = (0, 0, 4, 2).$$

Choosing the first move

Keep $x_2 = 0$ and increase x_1 from zero.

$$x_1 = t, \quad x_2 = 0, \quad p = 3t.$$

$$s_1 = 4 - t, \quad s_2 = 2 - t.$$

Profit increases, but both slacks must stay nonnegative.

How far can we move?

$$s_1 = 4 - t \geq 0 \quad \implies \quad t \leq 4.$$

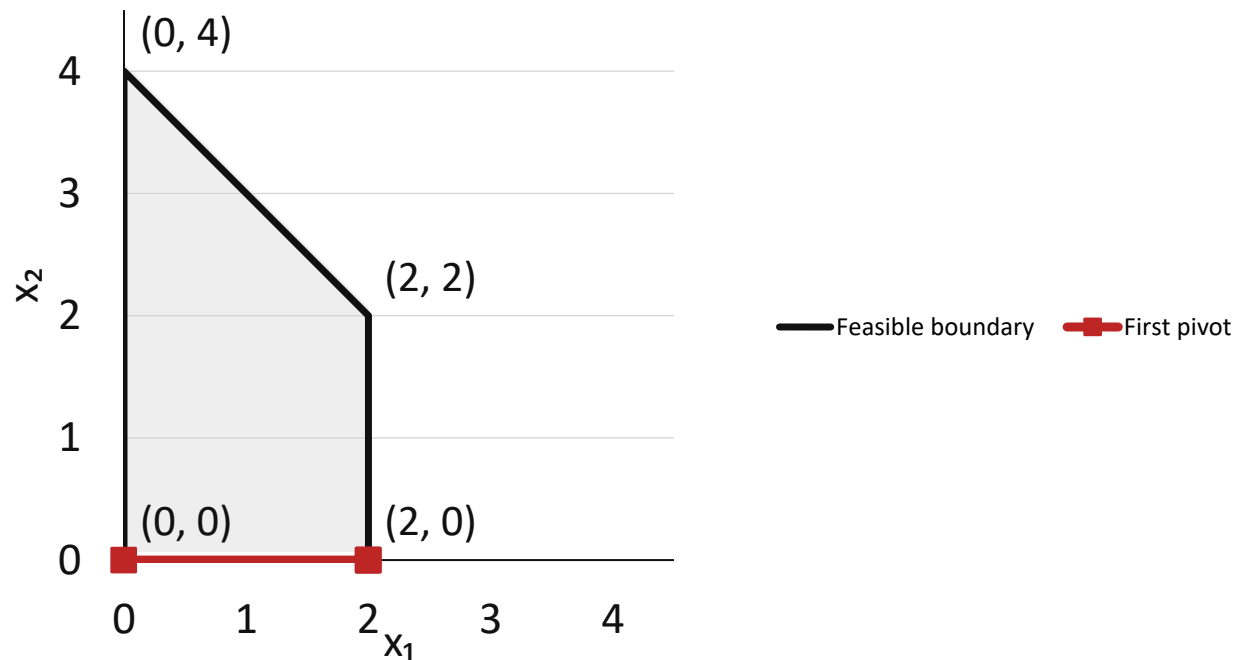
$$s_2 = 2 - t \geq 0 \quad \implies \quad t \leq 2.$$

$$t_{\max} = \min\{4, 2\} = 2.$$

The first constraint to become tight is $x_1 = 2$. This comparison is the idea behind the **ratio test**.

The first pivot

$$(0, 0) \longrightarrow (2, 0), \quad p : 0 \longrightarrow 6.$$



x_1 enters the basis and s_2 leaves: $s_2 = 0$.

The second move

Keep $s_2 = 0$, so $x_1 = 2$. Increase x_2 from zero.

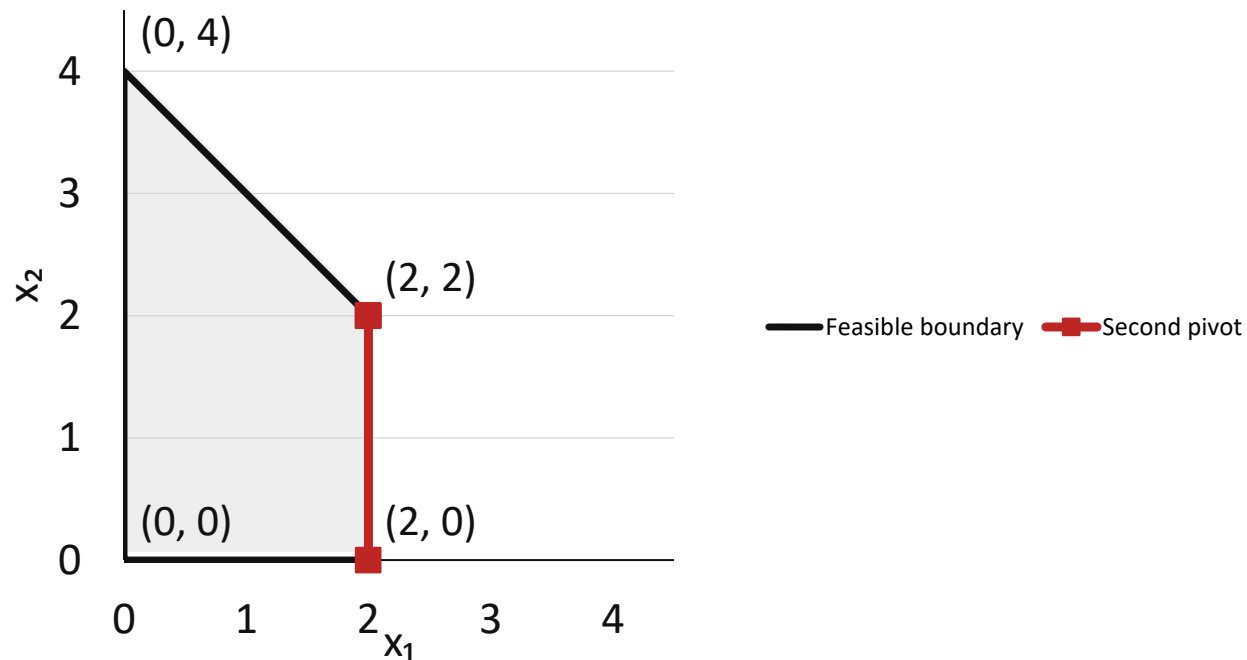
$$x_1 = 2, \quad x_2 = t, \quad p = 6 + 2t.$$

$$s_1 = 2 - t \geq 0 \quad \implies \quad t \leq 2.$$

Stop at $t = 2$, when all machine time has been used.

The second pivot

$$(2, 0) \longrightarrow (2, 2), \quad p : 6 \longrightarrow 10.$$



x_2 enters the basis and s_1 leaves: $s_1 = 0$.

Why can we stop?

Express production in terms of the slacks:

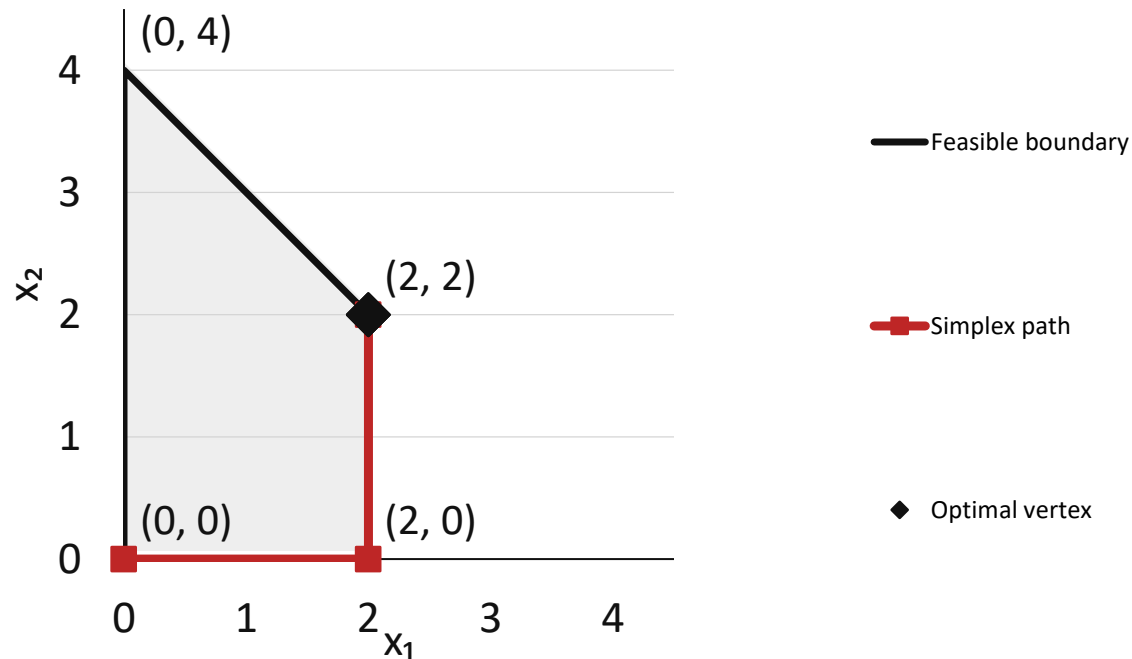
$$x_1 = 2 - s_2, \quad x_2 = 2 + s_2 - s_1.$$

$$p = 3x_1 + 2x_2 = 10 - 2s_1 - s_2 \leq 10.$$

Every feasible point has $s_1, s_2 \geq 0$. At $(2, 2)$ both slacks are zero, so the upper bound is attained.

The complete path

$$(0, 0) \longrightarrow (2, 0) \longrightarrow (2, 2).$$



Two pivots give profits 0, 6, and 10.

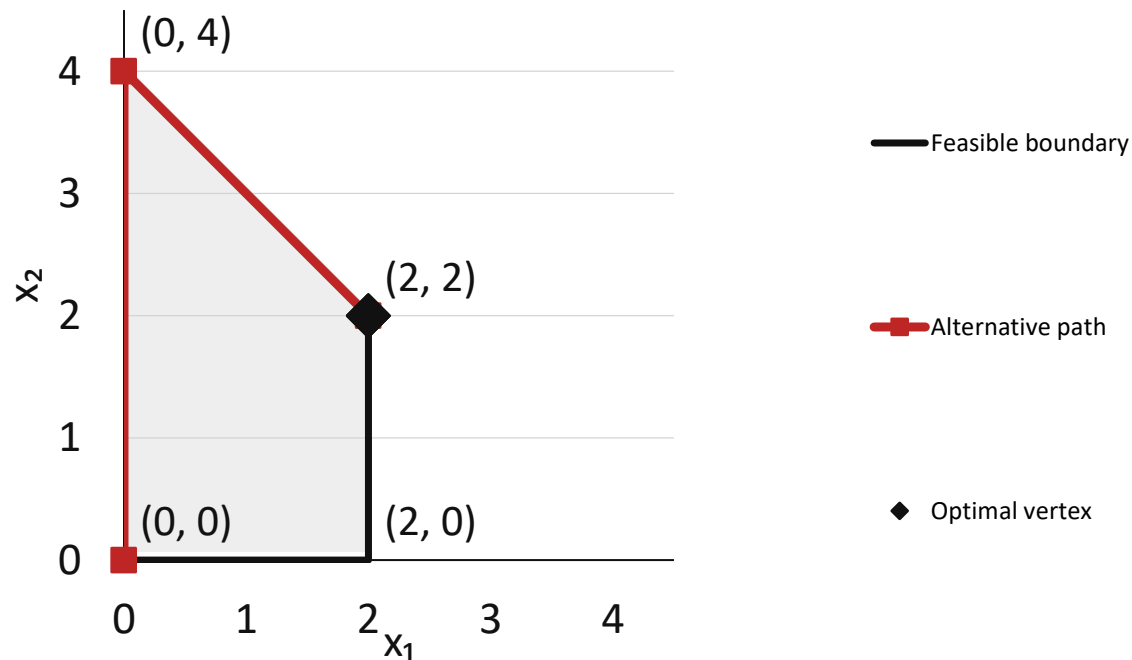
Simplex in a few steps

1. Start with a **feasible basis**.
2. Express the objective using the nonbasic variables. If every coefficient is nonpositive, stop: the current point is optimal.
3. Choose a variable with positive coefficient, keeping the other nonbasic variables at zero. Increase it until a basic variable reaches zero, then pivot and repeat.

If it can increase without limit while staying feasible, profit is unbounded.

A different valid route

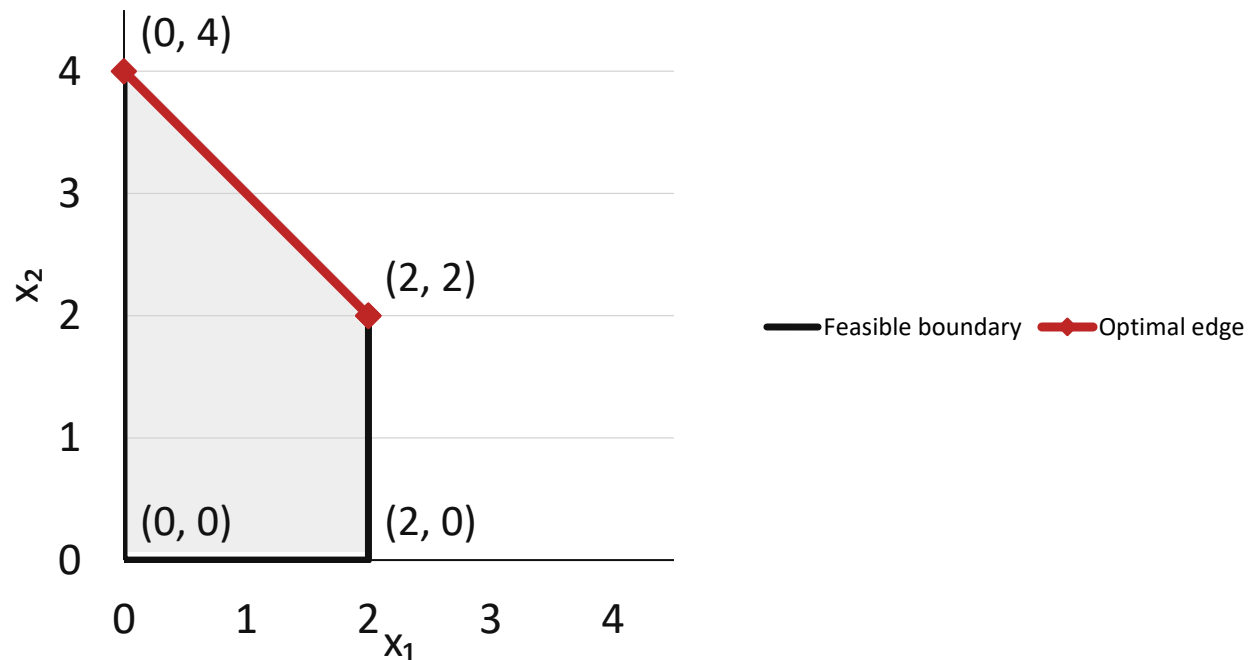
$$(0,0) \longrightarrow (0,4) \longrightarrow (2,2).$$



$$x_1 = t, x_2 = 4 - t \implies p = 8 + t, \quad 0 \leq t \leq 2.$$

Several optimal solutions

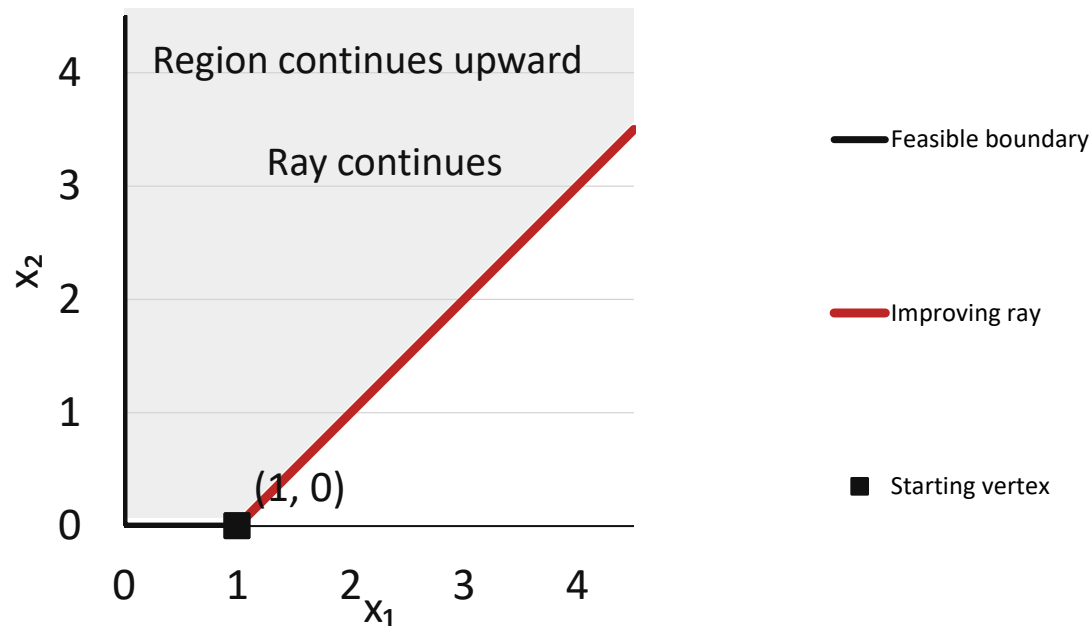
Change the objective to $p = 2x_1 + 2x_2$.



$$x_1 + x_2 = 4 \implies p = 8.$$

An improving unbounded ray

$$\max (x_1 + x_2) \quad \text{subject to } x_1 - x_2 \leq 1, \quad x_1, x_2 \geq 0.$$



$$(x_1, x_2) = (1 + t, t), \quad p = 1 + 2t \longrightarrow \infty \quad (t \geq 0).$$

Two practical details

Finding a start. Our example starts with nonnegative slacks. Other LPs may need an auxiliary feasibility problem, called Phase I.

Degeneracy. A basic variable may already be zero. A pivot can then change the basis without changing the point or profit.

Tie-breaking rules can prevent cycling.

An upper bound from constraints

Return to our original production LP. Multiply its machine constraint by 2 and its sales constraint by 1.

$$2(x_1 + x_2) \leq 8, \quad x_1 \leq 2.$$

$$\underbrace{2(x_1 + x_2) + x_1}_{3x_1 + 2x_2} \leq 8 + 2 = 10.$$

The weights $(2, 1)$ provide a certificate of optimality.

Choosing resource prices

Let $y_1, y_2 \geq 0$ be weights for machine time and sales capacity.

$$y_1(x_1 + x_2) + y_2x_1 \leq 4y_1 + 2y_2.$$

$$y_1 + y_2 \geq 3, \quad y_1 \geq 2.$$

These inequalities make the weighted resource use at least as large as the profit of every nonnegative production plan.

The dual linear program

Find the smallest upper bound produced by these weights:

$$\begin{array}{ll}\text{minimize} & 4y_1 + 2y_2 \\ \text{subject to} & y_1 + y_2 \geq 3, \\ & y_1 \geq 2, \\ & y_1, y_2 \geq 0.\end{array}$$

The production problem is the **primal**. This bound-finding problem is its **dual**.

Primal and dual side by side

Primal	$\max \longleftrightarrow \min$	Dual
$\max \quad c^T x$		$\min \quad b^T y$
s.t. $Ax \leq b,$		s.t. $A^T y \geq c,$
$x \geq 0.$		$y \geq 0.$

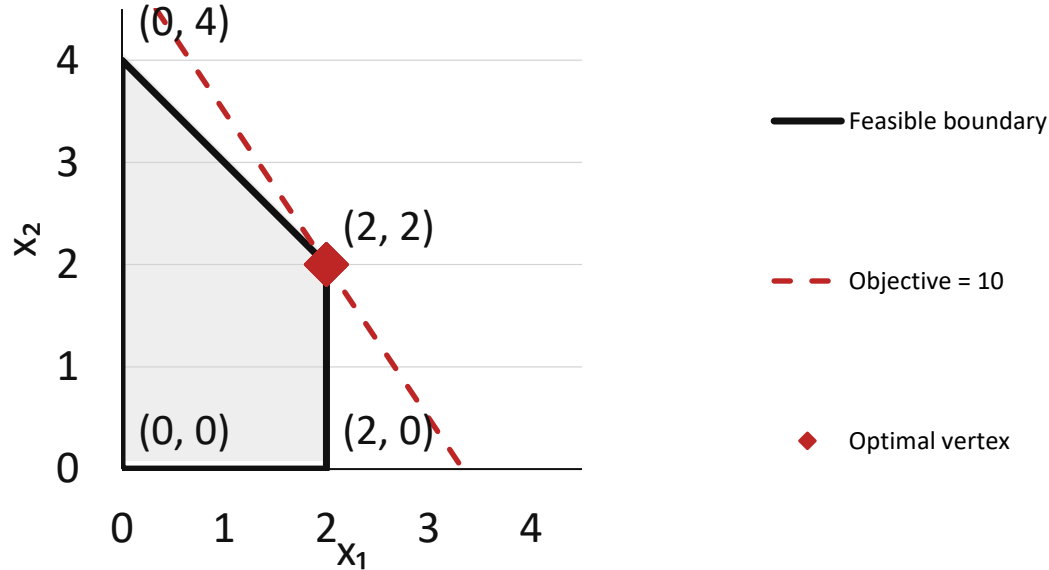
c : primal objective coefficients $\longrightarrow$ dual RHS

b : primal RHS $\longrightarrow$ dual objective coefficients

$A \longrightarrow A^T$: rows and columns exchange roles

Matching lower and upper bounds

$$\underbrace{3(2) + 2(2)}_{\text{feasible profit}} = 10 = \underbrace{4(2) + 2(1)}_{\text{dual bound}}.$$



Both solutions are optimal because their values match.

Weak duality

For every primal feasible x and dual feasible y ,

$$\begin{aligned} 3x_1 + 2x_2 &\leq (y_1 + y_2)x_1 + y_1x_2 \\ &= y_1(x_1 + x_2) + y_2x_1 \\ &\leq 4y_1 + 2y_2. \end{aligned}$$

A feasible dual value bounds **every** feasible primal value.

Reading and next lecture

Chong, Lu, and Žak, *An Introduction to Optimization*, fifth edition.

Simplex: Sections 16.1–16.4. Focus on the worked examples and the algorithm in Section 16.4.

Duality: Section 17.1 and the weak-duality discussion in Section 17.2.

Next: integer programming and combinatorial optimization.

Questions for discussion

1. At $(0, 4)$, neither coordinate can increase while the other stays fixed. Why is this point still not optimal for $3x_1 + 2x_2$?
2. Suppose a feasible production plan and feasible resource prices give the same value. Why does that prove optimality for both problems?